WEEK 4

1) In a distributed system, “clock skew” refers to:
A. Difference in clock frequencies (rates)
B. Relative difference in clock values at two processes
C. Total time taken by a message round trip
D. The number of clocks in the system

2) “Clock drift” refers to:
A. Relative difference in clock frequencies (rates)
B. Relative difference in clock values (timestamps)
C. Difference in CPU speeds of machines
D. The error in message timestamps only

3) Physical clocks are synchronized to an accurate real-time standard such as:
A. GPS-only time
B. UNIX epoch time
C. UTC (Universal Coordinated Time)
D. Lamport time

4) Which statement is correct?
A. Internal synchronization implies external synchronization
B. External synchronization implies internal synchronization with the same D
C. External synchronization with D implies internal synchronization with 2D
D. External and internal synchronization are identical concepts

5) A key problem with naive “ask time server and set clock” in an asynchronous system is:
A. Server time is always wrong
B. Message latency is unbounded, so error cannot be bounded
C. TCP cannot be used for time sync
D. Clocks cannot be adjusted forward

6) In Christian’s algorithm, the client measures:
A. Round-trip time (RTT)
B. One-way delay exactly
C. CPU utilization
D. Only server timestamp precision

7) Berkeley’s algorithm differs from Christian’s primarily because:
A. It synchronizes clocks to UTC directly
B. It uses a master that polls others and computes an average time
C. It requires hardware NIC changes
D. It works only for logical clocks

8) DTP (Datacenter Time Protocol) achieves bounded precision mainly by:
A. Using application-level retries
B. Using physical layer support of network devices
C. Storing timestamps on disk before replying
D. Running Paxos for every timestamp

9) The “happened-before” (→) relation forms a:
A. Total order over all events
B. Partial order over events
C. Random order over events
D. Strict FIFO order across all processes

10) In Lamport timestamps, on receiving a message with timestamp Tm, a process sets:
A. local = min(local, Tm)
B. local = local + Tm
C. local = max(local, Tm) + 1
D. local = Tm − 1